

So you have an **app directory wherein you're coding to build the web app** (with streamlit/python).

Let's say your main code for execution is called `start.py` and you have your Dockerfile outside of this directory.

A basic Dockerfile looks like :

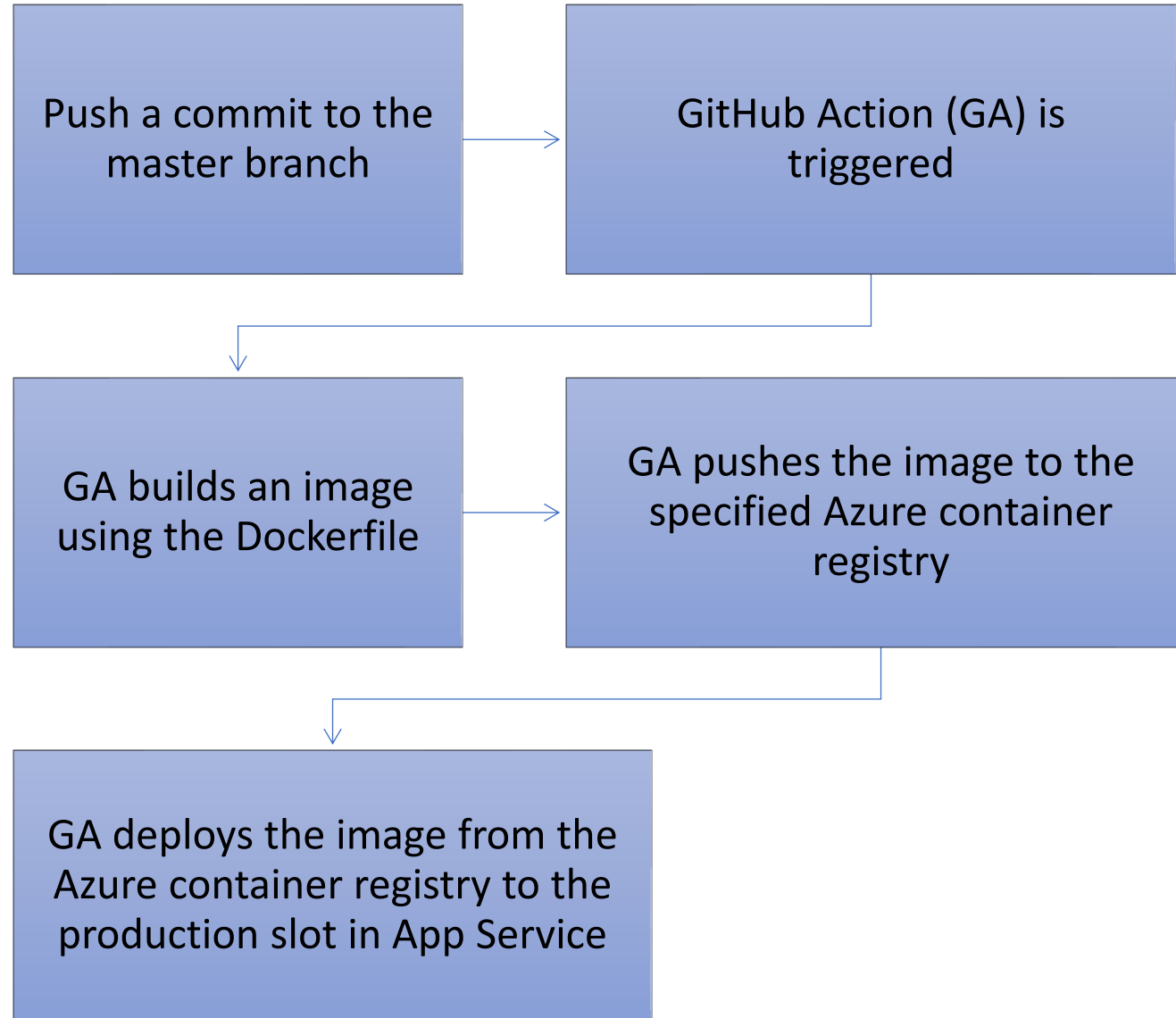
>> `pipenv run pip freeze > requirements.txt`

```
FROM python:3.9.1
EXPOSE 8501
COPY ./requirements.txt /requirements.txt
WORKDIR /
RUN pip3 install -r requirements.txt
COPY . /
ENTRYPOINT ["streamlit", "run"]
CMD ["start.py"]
WORKDIR /app
```

- **FROM** is what every Dockerfile begins with. It states the environment in which our image will be built and run, which in our case is the Python environment. This can also be, in other cases, an already existing docker image like that of an operating system such as Ubuntu, a server like nginx, and so on.
- **EXPOSE** indicates the port on which our app will be run.
- **COPY** command is used two times in the file. Its basic function is to copy over all the essential code and other files into the docker image. The **(dot)** indicates that we want to copy over the entire app directory into the docker image.
- **WORKDIR** is, as you might have already guessed, to state which is the working directory our app will be running from. We change it first to outside the app directory in order to run the next line → **pip3 install -r requirements.txt**. In the last line, we change it back to the app directory because the **start.py** file through which our app will execute is located there.
- Finally, **CMD** and **ENTRYPOINT** are two keywords used to describe a command to run on the terminal to start the app from within the container. The line to run our app is: **streamlit run start.py**. This can also be written as simply: **CMD ["streamlit", "run", "start.py"]**. There are many situations in which combining CMD and ENTRYPOINT would be the best solution for your Docker container. In such cases, **the executable is defined with ENTRYPOINT, while CMD specifies the default parameter,**

<https://www.docker.com/>

So you want to use **Azure App Services to deploy your app**. Here's how the Azure pipeline looks like :



CI/CD

